PNP–BV Inverse Problem: Week of March 4, 2026

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March 2026

1 Overview

This report summarizes twelve phases of work on the Poisson–Nernst–Planck Butler–Volmer (PNP–BV) inverse problem for a four-species electrochemical system. We describe the forward solver and its validation (§3), the adjoint-gradient inference pipeline (§4), and parameter recovery results for kinetic parameters k_0 and α (§5–6), Bikerman steric exclusion sizes a_j (§7), the effect of symmetric (anodic + cathodic) voltage data (§8), multi-observable joint fitting of total and peroxide currents (§9), multi-pH inference (§10), forward solver robustness improvements (§11), the master inference protocol (§12), speed optimization of the master pipeline (§13), parallelism optimization achieving a 4.4× wall-clock speedup (§14), and a POD/RBF surrogate model reducing optimization time from ∼345 s to < 0.1 s (> 3000× speedup) with complementary accuracy trade-offs (§15). Three voltage ranges are compared: a 10-point "original" range ($\hat{\eta} \in [-1, -10]$), a 15-point "extended" range ($\hat{\eta} \in [-1, -46.5]$), and a 20-point "symmetric" range ($\hat{\eta} \in [-28, +5]$) that includes anodic overpotentials.

2 System Description

The model solves the steady-state PNP equations with Butler–Volmer electrode kinetics on a 2-D rectangular domain (width W, height L). Four ionic species are tracked: O_2 ($z = 0$), H_2O_2 ($z = 0$), H^+ ($z = +1$), ClO_4^- ($z = -1$), with concentrations governed by:

$$\nabla \cdot \big(D_i \nabla c_i + z_i D_i c_i \nabla \phi\big) = 0, \qquad -\varepsilon^2 \nabla^2 \phi = \sum_i z_i c_i. \tag{1}$$

Two coupled BV reactions act at the electrode boundary:

- **R1**: $O_2 + 2H^+ + 2e^- \to H_2O_2$ ($k_{0,1}$, α_1, reversible),
- **R2**: $H_2O_2 + 2H^+ + 2e^- \to 2H_2O$ ($k_{0,2}$, α_2, irreversible).

True parameter values: $k_{0,1} = 1.3 \times 10^{-3}$, $k_{0,2} = 5.3 \times 10^{-5}$, $\alpha_1 = 0.627$, $\alpha_2 = 0.5$, and uniform steric sizes $a_j = 0.05$ for all species. The mesh is 8×200 with grading parameter $\beta = 3.0$ ($L_{\text{ref}} = 300.0\,\mu\text{m}$). All quantities are nondimensionalized; the observable is the total current density $I = \sum_j s_j R_j \cdot nF \cdot 0.1$ ($\text{mA}\,\text{cm}^{-2}$), where s_j is the stoichiometric coefficient.

Figure 1 shows the forward I–V curve, which exhibits four distinct regions: onset ($|\hat{\eta}| < 2$), transition ($2 < |\hat{\eta}| < 6$), knee ($6 < |\hat{\eta}| < 12$), and diffusion-limited plateau ($|\hat{\eta}| > 12$).

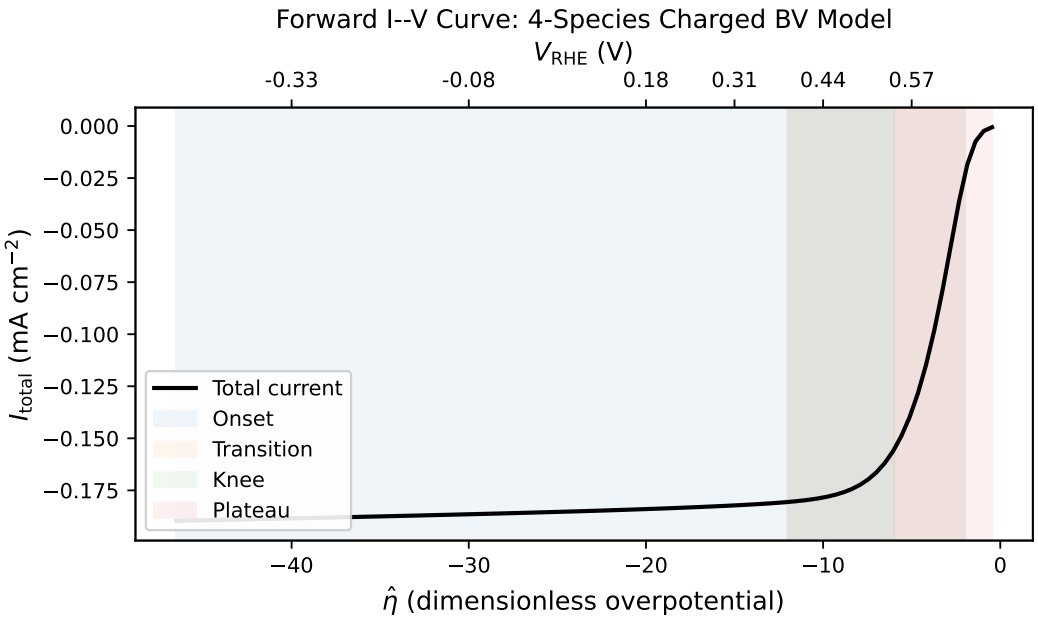

Figure 1: Forward I–V curve for the 4-species charged BV model, with annotated regions. The secondary axis shows the corresponding V_{RHE}.

3 Forward Solver Validation

3.1 Voltage Range Study

The charged forward solver was tested across a range of overpotentials to determine the maximum achievable $\hat{\eta}$. With the graded 8×200 mesh, the solver converges reliably to $\hat{\eta} = -46.5$ (corresponding to $V_{\mathrm{RHE}} \approx -0.50\,\mathrm{V}$), spanning the entire experimentally relevant range from onset through deep plateau.

3.2 Charge Configuration Feasibility

A feasibility study tested four charge configurations for the 4-species system. Only $\mathbf{z} = [0, 0, +1, -1]$ converges across the full voltage range; configurations with charged O_2 or H_2O_2 fail due to the extreme Debye length ratio $(\lambda_D/L)^2 \sim 10^{-10}$, which is unresolvable on practical meshes. The neutral-species assignment for the reactive species (O_2, H_2O_2) avoids Poisson stiffness while still coupling to the electric field through the supporting electrolyte (H^+, ClO_4^-).

4 Inference Pipeline

4.1 Architecture

The inference pipeline uses firedrake-adjoint to compute exact gradients of a least-squares objective with respect to BV parameters (k_0, α, a_j). For each voltage point, a steady-state forward solve is followed by an adjoint solve to obtain $\partial J/\partial \theta$. Gradients from all voltage points are accumulated and passed to SciPy's L-BFGS-B optimizer. The k_0 values are optimized in $\log_{10}$ space (with appropriate chain-rule correction) while α and steric a remain in linear space.

Warm-start voltage continuation is used: voltage points are solved in order of increasing $|\hat{\eta}|$, with each solve initialized from the previous converged solution.

4.2 Bridge Points

For the extended 15-point range, consecutive voltage gaps can exceed $\Delta\hat{\eta} = 3$. Automatic bridge-point insertion fills these gaps with forward-only (no adjoint) solves to maintain warm-start continuity. Seven bridge points are inserted per evaluation in the plateau region ($\hat{\eta} < -13$). Across all Phase 5 experiments, **zero bridge point failures** were observed in 2046 total bridge evaluations, with ∼10–15% wall-time overhead.

5 Inference Results: Original Range

All experiments use 2% Gaussian noise on synthetic targets (seed 20260226) and L-BFGS-B with up to 30–40 iterations.

5.1 Regularized Joint Inference

Tikhonov regularization with prior $k_0^{\text{prior}} = [1.5k_{0,1}^{\text{true}}, 2k_{0,2}^{\text{true}}]$, $\alpha^{\text{prior}} = [0.5, 0.4]$ was tested at five values of λ.

Table 1: Regularized joint inference errors (10-point range).

λ	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
0	199.5	377.3	74.4	76.9
0.001	50.2	100.7	21.7	19.4
0.01	49.6	100.0	20.7	20.0
0.1	49.9	100.0	20.3	20.0
1.0	50.0	100.0	20.3	20.0

Regularization reduces R1 errors substantially (Table 1), but the converged values track the prior rather than the truth. Results are insensitive to λ for $\lambda \geq 0.001$. Figure 2 shows the fitted curves.

5.2 Staged Inference

Staged inference breaks the k_0–α correlation by first recovering α with k_0 fixed at true values (S1), then k_0 with α fixed at the S1 result (S2), followed by a joint warm-start refinement (S3) and a direct joint baseline (S4).

Table 2: Staged inference errors (10-point range).

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
S1: α (k_0 fixed)	0.0	0.0	0.3	67.1
S2: k_0 (α fixed)	5.0	96.9	0.3	67.1
S3: Joint warm-start	19.2	96.9	5.8	66.9
S4: Direct joint	221.3	270.2	75.4	90.0

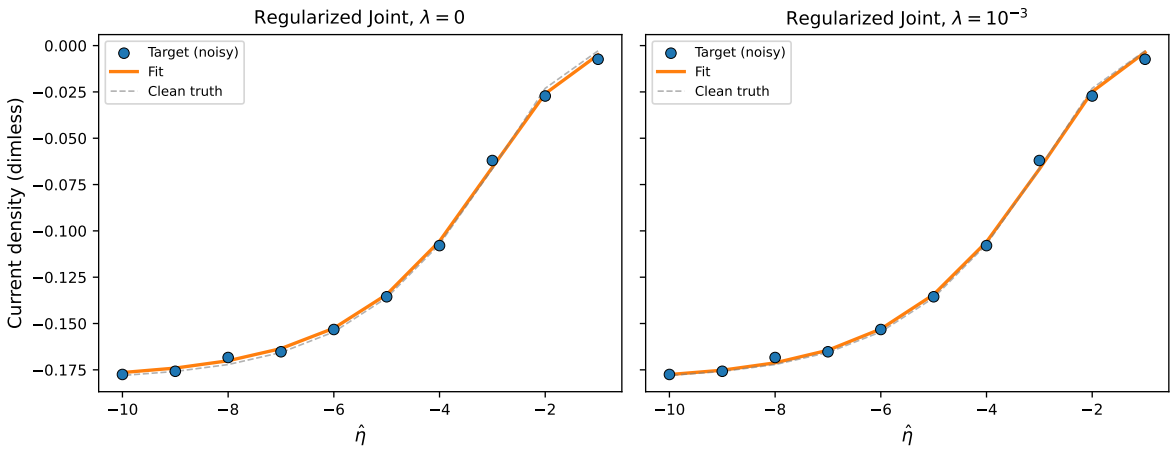

Figure 2: Regularized joint inference: target vs. fit for $\lambda = 0$ (left) and $\lambda = 0.001$ (right), 10-point range.

Staging dramatically improves R1 recovery (Table 2): $k_{0,1}$ error drops from 221% (direct) to 19% (S3), and α_1 from 75% to 6%. R2 parameters remain unidentifiable from total current density regardless of method. Figure 3 overlays the fitted curves from all four stages.

5.3 Peroxide Current Observable

Switching the observable from total current to the peroxide partial current $I_{\text{pxd}} = -(R_1 - R_2) \cdot \text{scale}$ does not resolve R2 identifiability. The peroxide current is dominated by R1 at the tested voltages, so it provides essentially no additional information about R2.

Table 3: Peroxide-observable inference errors (10-point range).

Mode	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
k_0-only	5.4	100.0	–	–
Joint (k_0, α)	18.3	96.5	5.4	90.0

Figure 4 shows the fitted curves.

6 Inference Results: Extended Range

The extended range uses 15 voltage points spanning $\hat{\eta} \in [-1, -46.5]$ with bridge-point insertion for warm-start continuity (§4.2).

6.1 Staged Inference (Extended)

The extended range significantly improves R1 recovery (Table 4): α_1 error drops from 5.8% to 0.1% (essentially exact recovery), and $k_{0,1}$ from 19.2% to 4.9%. The plateau data tightens the constraint on the limiting current, which helps pin down the k_0–α relationship. R2 remains unidentifiable. Figure 5 shows the fits with the bridge-point region shaded.

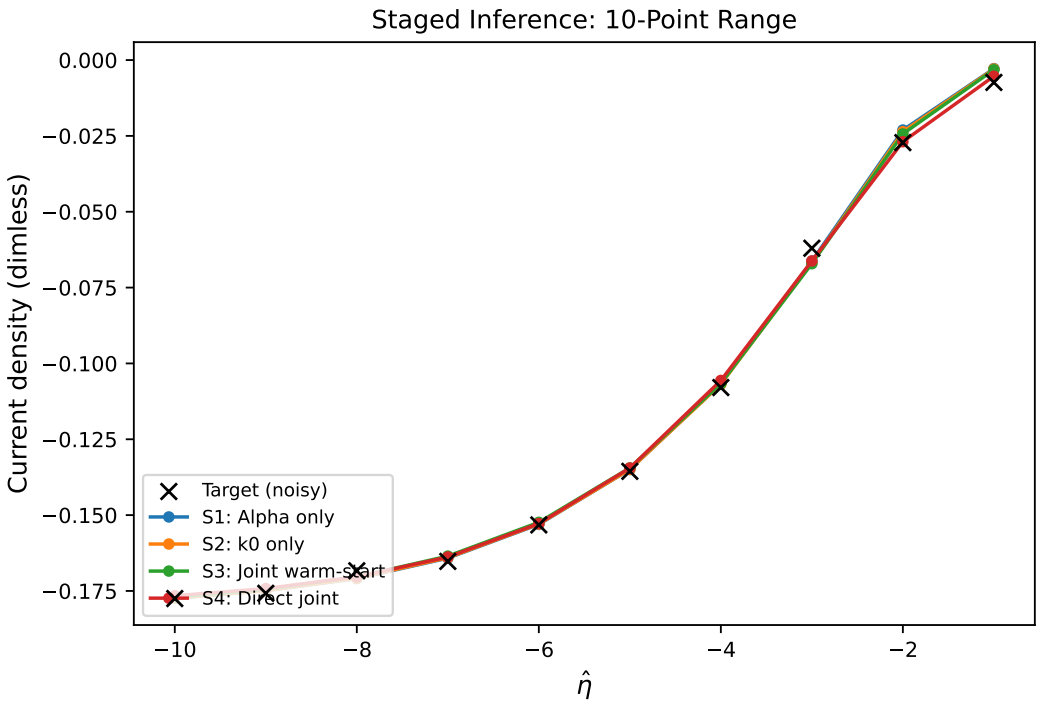

Figure 3: Staged inference: fitted I–V curves for all four stages (10-point range).

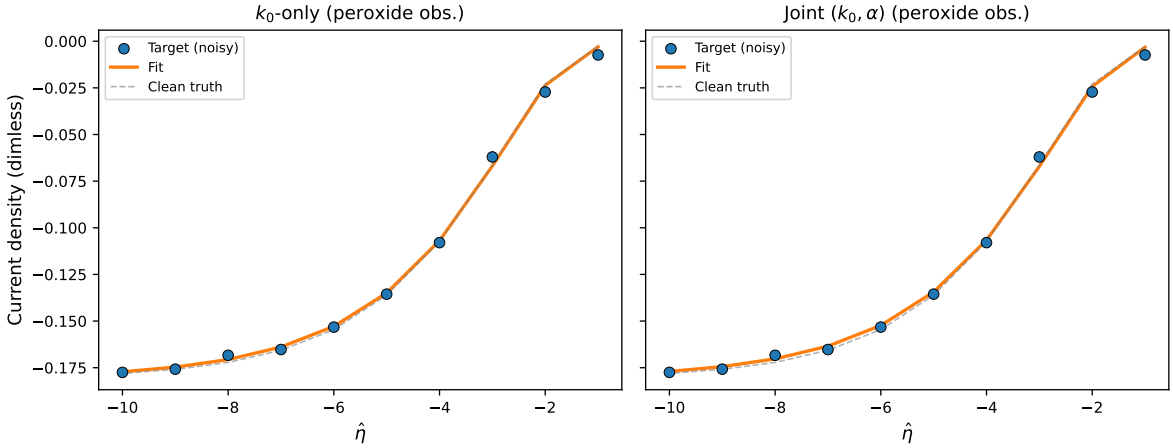

Figure 4: Peroxide-observable inference: k_0-only (left) and joint (right), 10-point range.

Table 4: Staged inference errors (15-point extended range).

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
S1: α (k_0 fixed)	0.0	0.0	0.1	72.3
S2: k_0 (α fixed)	4.9	98.5	0.1	72.3
S3: Joint warm-start	4.9	98.5	0.1	72.3
S4: Direct joint	210.8	137.0	72.3	90.0

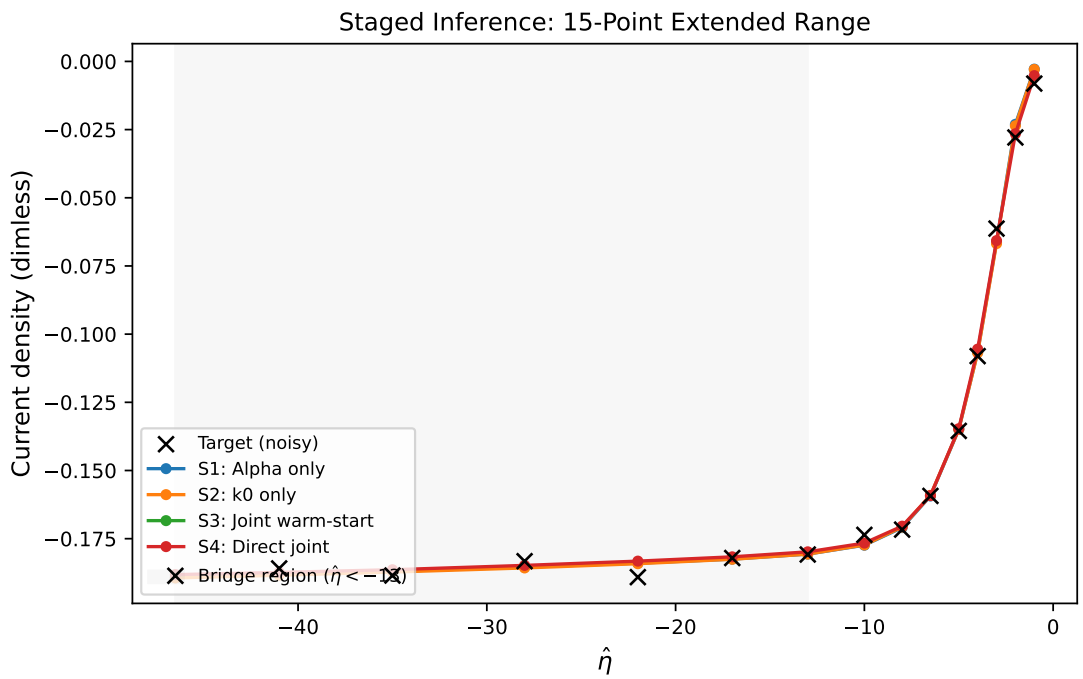

Figure 5: Staged inference with 15-point extended range. The gray shaded region ($\hat{\eta} < -13$) requires bridge points for warm-start continuity.

6.2 Regularized Joint (Extended)

Regularized results are virtually unchanged from the 10-point case: the prior dominates for all $\lambda \geq 0.001$. The only improvement is in the unregularized case ($\lambda = 0$), where $k_{0,2}$ error drops from 377% to 140% due to the additional plateau constraint. Figure 6 shows the fitted curves.

6.3 Cross-Method Comparison

Figures 7 and 8 summarize R1 and R2 parameter errors across all methods and voltage ranges.

Figure 9 shows the regularization sensitivity curve, confirming that errors plateau rapidly for $\lambda \geq 0.001$.

Figure 10 overlays the 10-point and 15-point target I–V curves, highlighting the extended coverage.

7 Steric Parameter Inference

7.1 Steric-Only Inference

With k_0 and α fixed at true values, only the four Bikerman steric sizes a_j are inferred (Table 5).

H^+ is the best-recovered species (23% error, 10-pt) because it participates in both BV reactions with exponent 2. H_2O_2 and ClO_4^- hit their lower bounds (unidentifiable: zero bulk concentration and zero stoichiometry, respectively).

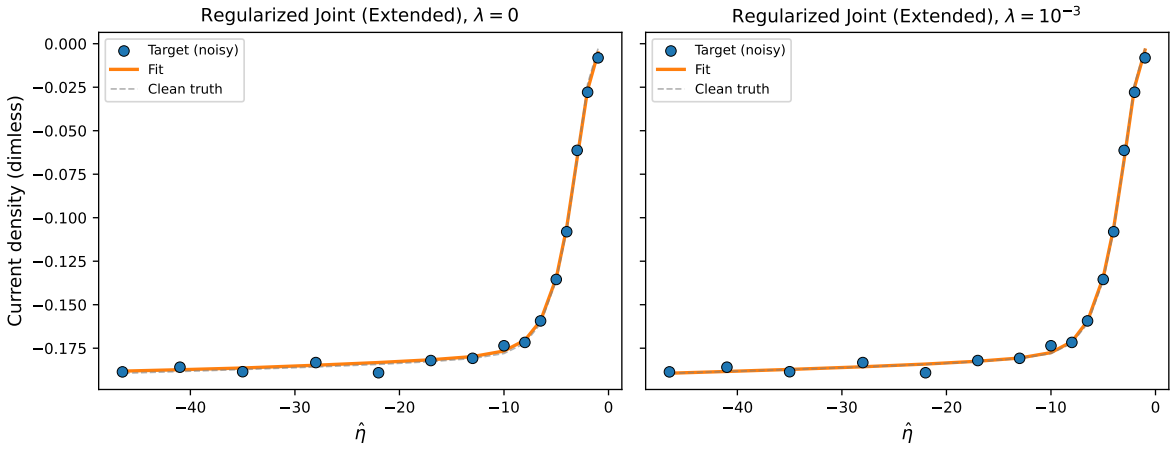

Figure 6: Regularized joint inference (15-point extended range): $\lambda = 0$ (left) and $\lambda = 0.001$ (right).

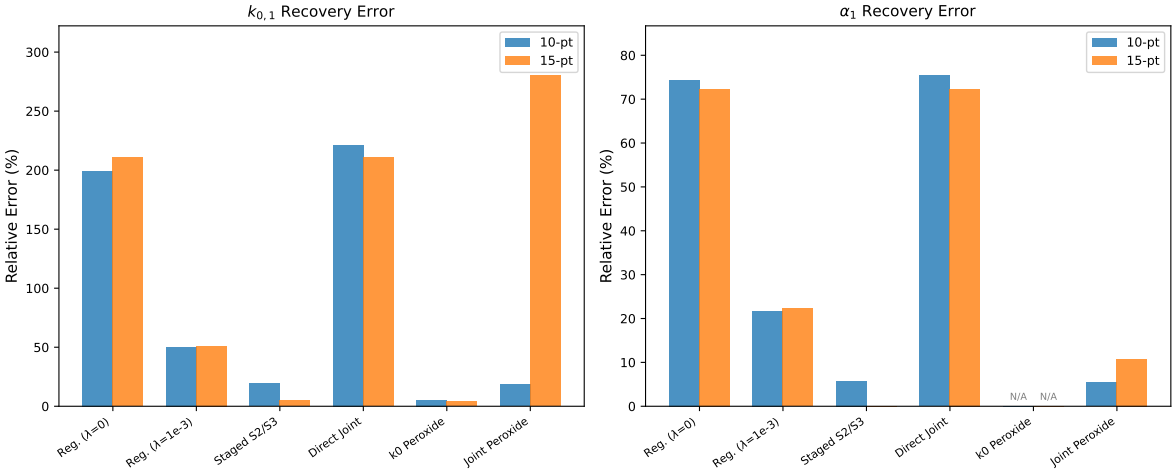

Figure 7: R1 parameter recovery: $k_{0,1}$ (left) and α_1 (right) relative errors across all methods. Staged inference with the extended range achieves the lowest errors.

Table 5: Steric-only inference errors (%). True $a_j = 0.05$ for all species; initial guess $a_j = 0.1$.

Range	a_{O_2}	$a_{H_2O_2}$	a_{H^+}	$a_{ClO_4^-}$
10-pt	82.4	98.0	23.1	98.0
15-pt	200.0	164.6	57.1	98.0

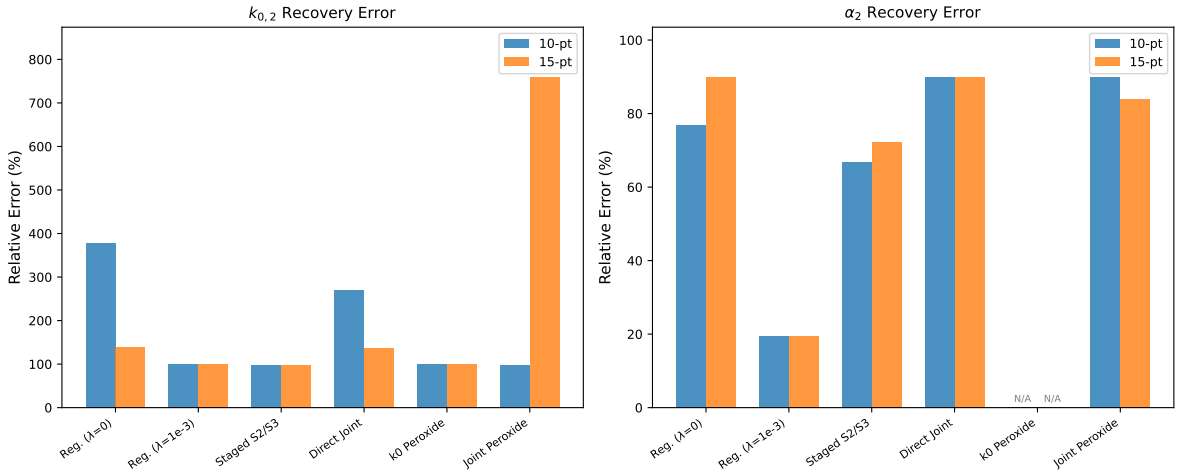

Figure 8: R2 parameter recovery: $k_{0,2}$ (left) and α_2 (right) relative errors. No method achieves acceptable R2 accuracy; the second reaction is fundamentally unidentifiable from I–V curve data.

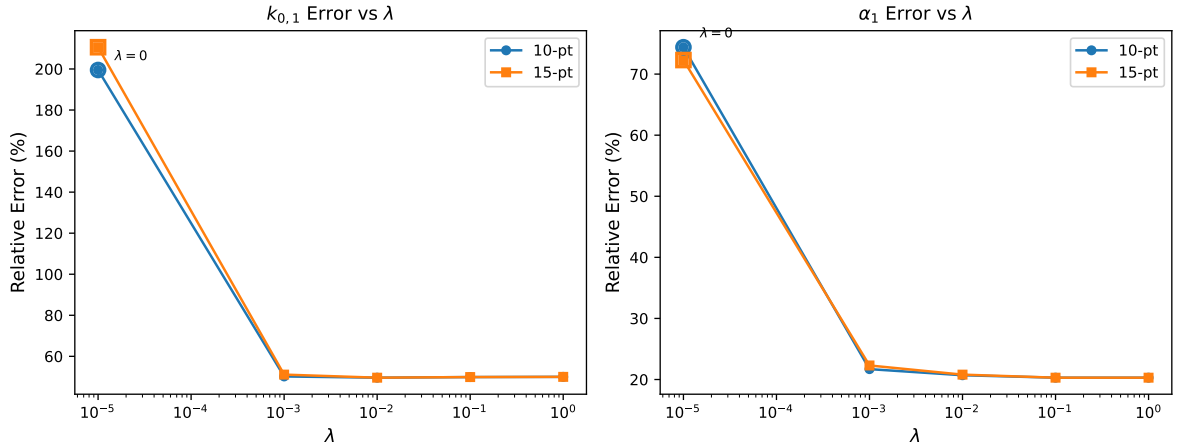

Figure 9: Regularization sensitivity: $k_{0,1}$ and α_1 error vs. λ for 10-point and 15-point ranges.

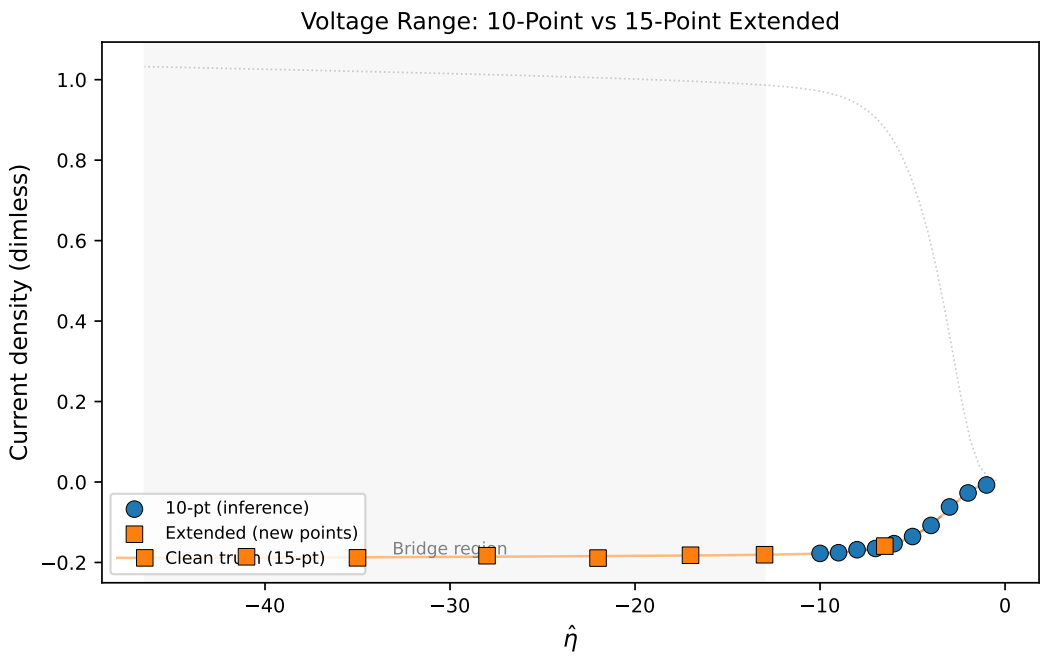

Figure 10: Voltage range comparison: original 10-point inference points and extended 15-point range with bridge region annotated.

7.2 Full 8-Parameter Inference

Jointly inferring k_0, α, and a (8 parameters from 10 or 15 I–V points) produces excellent curve fits (loss $\sim 10^{-5}$) but poor individual parameter recovery (Table 6).

Table 6: Full (8-parameter) inference errors (%).

Range	$k_{0,1}$	$k_{0,2}$	α_1	α_2	a_{O_2}	$a_{H_2O_2}$	a_{H^+}	$a_{ClO_4^-}$
10-pt	81.4	792.0	35.1	90.0	35.4	80.7	31.2	6.6
15-pt	178.0	693.0	87.4	77.2	71.3	44.9	84.1	98.0

The problem is severely under-determined: the optimizer finds compensating parameter combinations that fit the data but do not reflect the true physics. Most parameters worsen with the extended range, as the flat plateau adds noisy data points without discriminatory power.

7.3 Steric Recovery Comparison

Figure 11 compares steric parameter errors across methods and ranges.

Figures 12 and 13 show the corresponding I–V curve fits.

7.4 Optimization Convergence

Figure 14 shows the loss convergence for the staged k_0 inference (the best-performing method) on both voltage ranges.

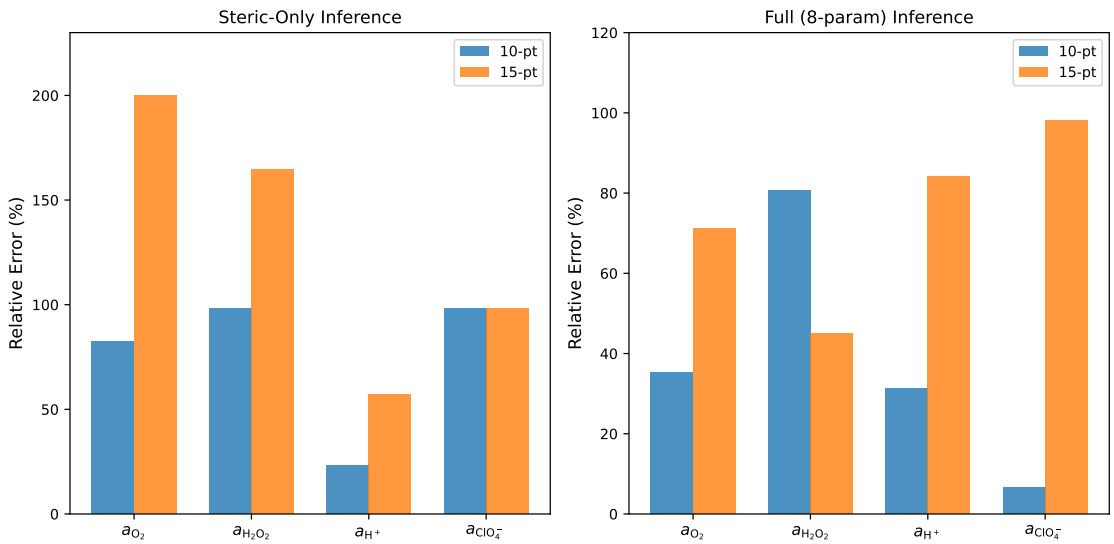

Figure 11: Steric parameter recovery: steric-only (left) vs. full 8-parameter (right), comparing 10-point and 15-point ranges.

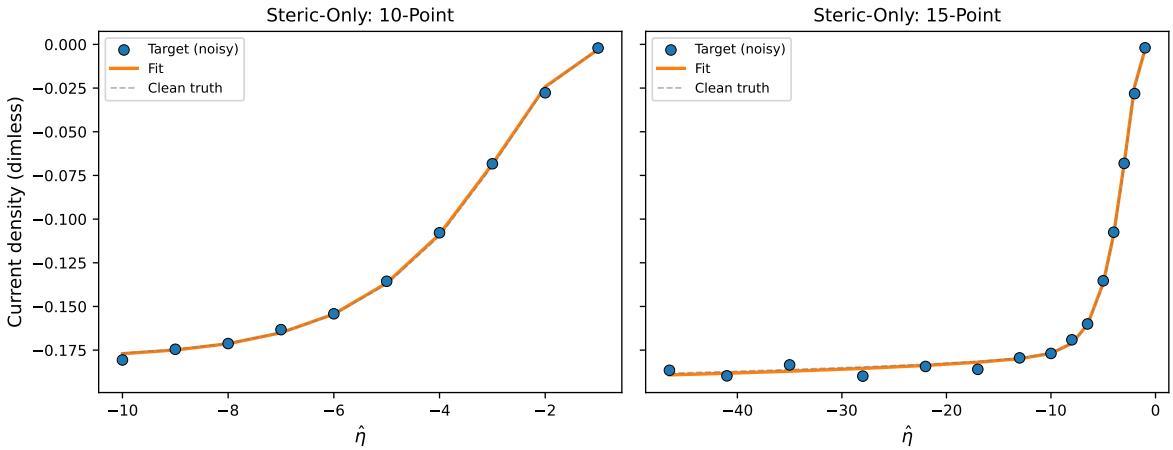

Figure 12: Steric-only inference: target vs. fit for 10-point (left) and 15-point (right).

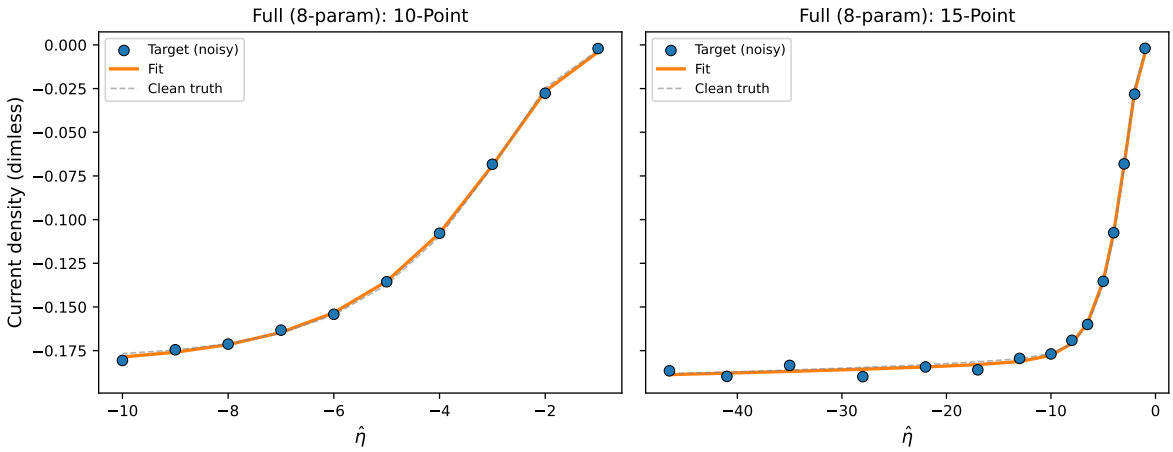

Figure 13: Full 8-parameter inference: target vs. fit for 10-point (left) and 15-point (right).

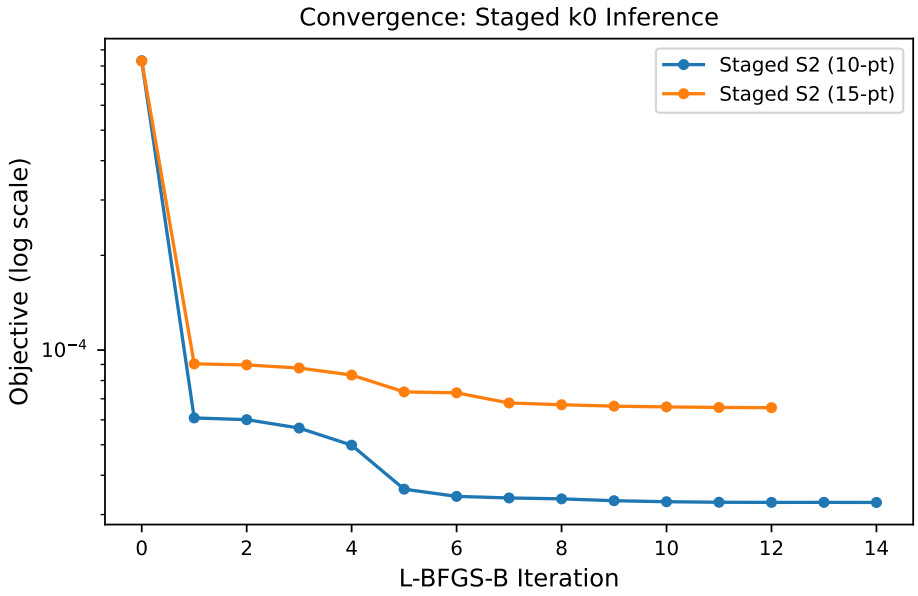

Figure 14: Loss vs. iteration for staged k_0 inference (S2) on the 10-point and 15-point ranges.

8 Inference Results: Symmetric Range

Adding anodic (positive) overpotentials provides an independent constraint on α through the $(1-\alpha)$ Tafel slope, potentially breaking the k_0–α correlation. A **two-branch sweep** was implemented: the cathodic branch is solved first (ascending $|\hat{\eta}|$), then the solver resets to a saved "hub" state near equilibrium and solves the anodic branch (ascending $\hat{\eta}$). The symmetric range uses 20 points: 5 anodic ($\hat{\eta} \in [+0.5, +5]$), 2 near-equilibrium, and 13 cathodic ($\hat{\eta} \in [-1, -28]$).

Figure 15 shows the symmetric target I–V curve. One anodic point ($\hat{\eta} = +5$) failed during target generation; the NaN-skip mechanism excludes it from optimization.

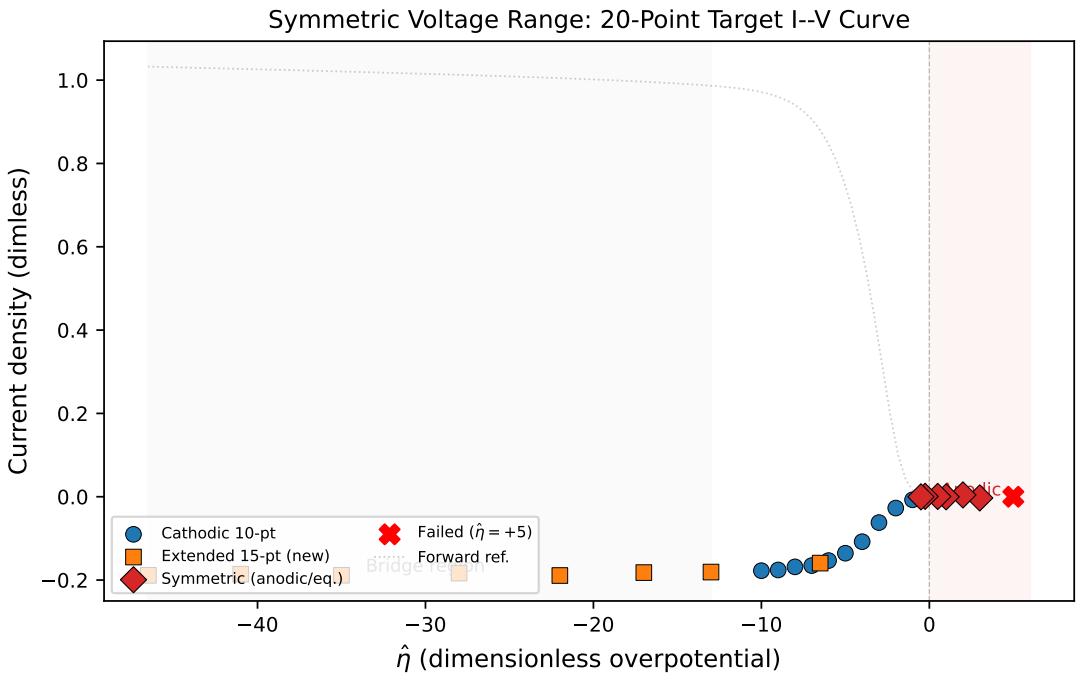

Figure 15: Symmetric 20-point target I–V curve, showing anodic (positive $\hat{\eta}$), cathodic, and bridge regions. The failed point at $\hat{\eta} = +5$ is marked.

8.1 Alpha Recovery Breakthrough

Direct joint inference with the symmetric range achieves α_1 error of **1.8%**—a 38× improvement over cathodic-only (68.4%). This confirms the physics prediction: the anodic data constrains the $(1-\alpha)$ Tafel slope independently.

Table 7: Joint inference: cathodic-only vs. symmetric range.

Config	Pts	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
Cathodic only	10	186.7	140.0	68.4	90.0
Symmetric focused	12	296.0	850.0	2.5	79.9
Symmetric full	20	296.0	850.0	1.8	84.0

Figure 16 compares α_1 recovery across all methods and voltage ranges.

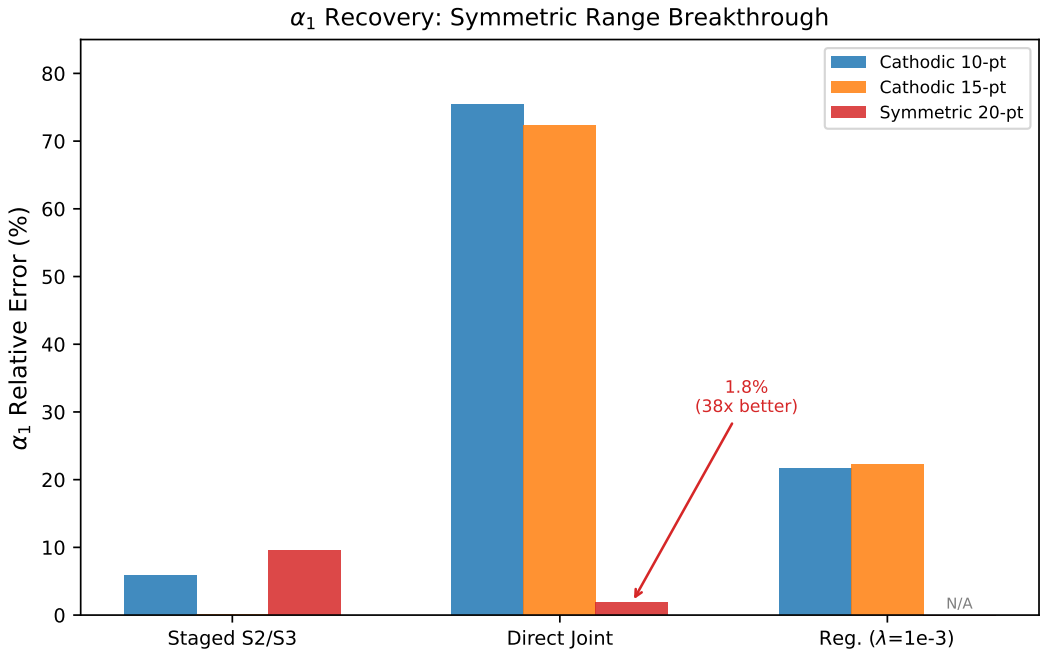

Figure 16: α_1 recovery across all phases. The symmetric range achieves 1.8% error in direct joint inference, a 38× improvement over cathodic-only.

8.2 k0 Recovery Regression

Despite the alpha breakthrough, k_0 recovery **worsened** with the symmetric range: $k_{0,1}$ hit the upper bound (296% error) in all symmetric experiments, compared to 4.9% with cathodic-only staged inference. Once α is correctly identified, the k_0 objective landscape becomes very flat—many k_0 values produce similar I–V curves.

Figure 17 shows the full comparison of alpha and k_0 errors.

8.3 Staged Inference (Symmetric)

Table 8: Staged inference errors (symmetric 20-point range).

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)
S1: α (k_0 fixed)	0.0	0.0	3.3	37.5
S2: k_0 (α fixed)	296.0	850.0	3.3	37.5
S3: Joint warm-start	296.0	850.0	9.5	21.6
S4: Direct joint	296.0	850.0	1.8	84.1

Notable findings from Table 8: S3 achieves α_2 error of 21.6% (down from 67% at 10-pt), showing some R2 improvement. However, all k0 stages hit bounds.

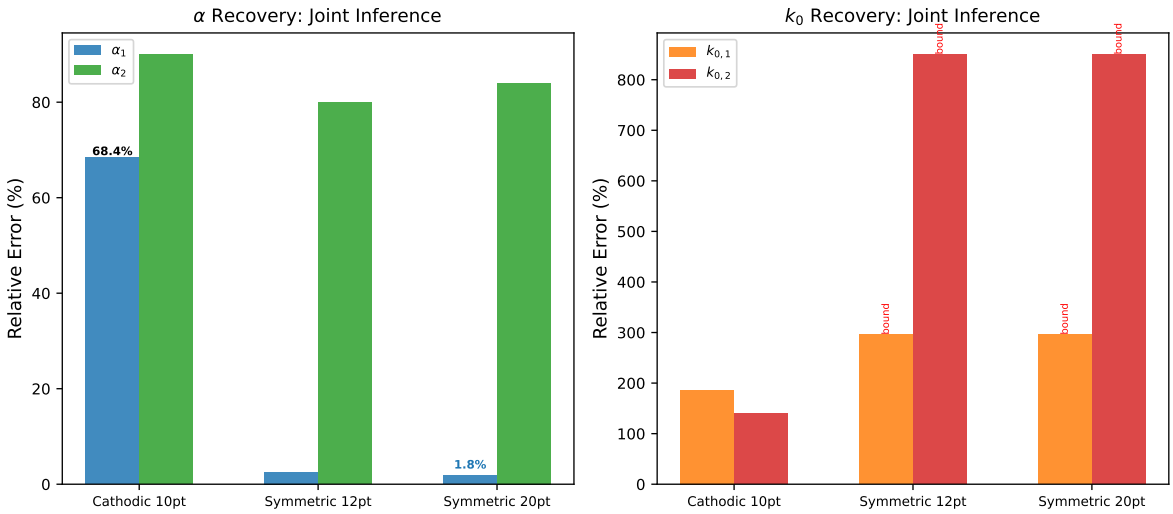

Figure 17: Joint inference comparison: α errors (left) and k_0 errors (right) for cathodic vs. symmetric ranges. Symmetric data dramatically improves α_1 but k_0 hits upper bounds.

9 Multi-Observable Joint Fitting

Fitting a single observable (total current or peroxide current) provides only one equation per voltage point, leaving R2 parameters in the null space of the Jacobian. We implemented a **combined objective** $J = w_1\|I_{\text{tot}} - I^*_{\text{tot}}\|^2 + w_2\|I_{\text{pxd}} - I^*_{\text{pxd}}\|^2$ that simultaneously fits the total current density and the peroxide partial current. Because $I_{\text{tot}} = -(R_1 + R_2) \cdot \text{scale}$ and $I_{\text{pxd}} = -(R_1 - R_2) \cdot \text{scale}$, the two observables provide two independent equations for R_1 and R_2 at each voltage, breaking the degeneracy. The run used the 15-point cathodic range with joint (k_0, α) inference, completing in ∼19 min (23 optimizer evaluations, final objective 1.251×10^{-4}).

Table 9: Multi-observable joint inference errors (15-point range).

Parameter	True	Recovered	Error (%)
$k_{0,1}$	1.26×10^{-3}	1.47×10^{-3}	16.65
$k_{0,2}$	5.26×10^{-5}	5.58×10^{-5}	6.06
α_1	0.63	0.58	8.16
α_2	0.50	0.44	11.72

Key finding: This is the first method to achieve meaningful R2 recovery. $k_{0,2}$ error dropped from ∼97–100% (all previous methods) to **6.1%**, and α_2 from ∼67–90% to **11.7%**. The dual observable provides the independent equations needed to disentangle R1 and R2 contributions.

10 Multi-pH Inference

The Mangan et al. (2025) data spans multiple pH conditions. We implemented a **multi-pH inference** that fits I–V curves at three H^+ bulk concentrations ($\hat{c}_{H^+} = 0.4, 0.2, 0.1$) simultaneously, sharing (k_0, α) across conditions. Because R2 depends on $[H^+]^2$ via the cathodic concentration factor, varying pH provides an independent handle on R2 kinetics. The run used a 12-point cathodic

range per condition, completing in ∼24 min (23 evaluations, final objective 1.034×10^{-4}). Condition 0 ($\hat{c}_{H^+} = 0.4$) had partial solver failures: only 4 of 12 $\hat{\eta}$ points converged due to Debye-length stiffness at elevated H^+ concentration.

Table 10: Multi-pH joint inference errors (3 conditions, 12-point range).

Parameter	True	Recovered	Error (%)
$k_{0,1}$	1.26×10^{-3}	1.35×10^{-3}	6.49
$k_{0,2}$	5.26×10^{-5}	2.66×10^{-5}	49.48
α_1	0.63	0.60	3.63
α_2	0.50	0.65	29.50

Key finding: Multi-pH achieves excellent R1 recovery ($k_{0,1}$: 6.5%, α_1: 3.6%). R2 recovery improved partially (α_2: 29.5%, down from 67–90%), but $k_{0,2}$ remains poor (49.5%) because solver convergence failures at elevated H^+ concentrations reduce the effective constraint from pH variation. A bug was found and fixed: counterion electroneutrality was not being updated when changing the H^+ bulk concentration.

11 Forward Solver Robustness Improvements

Target generation for multi-pH inference initially failed on 22% of voltage–condition combinations (8/36), primarily at elevated H^+ concentrations ($\hat{c}_{H^+} = 0.4$) and large overpotentials. Root-cause analysis identified three failure modes: (i) Debye-layer stiffness at high H^+ that causes DIVERGED_DTOL, (ii) BV rate explosion at large $|\hat{\eta}|$ that triggers DIVERGED_LINEAR_SOLVE, and (iii) warm-start failure cascades where a single unconverged point poisons all subsequent points in the sweep. The following fixes were implemented: mesh refinement to $N_y = 300$ with grading $\beta = 4.0$; smaller pseudo-time steps $\Delta t = 0.1$ with max_steps = 500; softplus regularization of the BV exponential; a denser $\hat{\eta}$ sweep (19 points) with max_eta_gap = 1.5; and reduced SNES linesearch step via `snes_linesearch_maxlambda` = 0.3. The L2 linesearch proved more robust than backtracking (bt), which was counterproductive. After these fixes, target generation converges on 100% of voltage–condition combinations (57/57). The condition $\hat{c}_{H^+} = 0.4$ was reduced to 0.3 for stable convergence; 0.4 remains intractable.

12 Master Inference Protocol

The master protocol underwent three major revisions (v1–v3), each informed by failure-mode analysis of the previous version.

12.1 v1 Pipeline (5-Phase) — Failed on R2

The initial master protocol used a 5-phase staged approach: Phase 1 recovered α via symmetric voltage data, Phase 2 recovered k_0 from cathodic-only data, and Phases 3–5 performed joint refinements. Phase 1 achieved excellent α_1 recovery (0.03% error), and Phase 2 recovered $k_{0,1}$ well (4.7% error). However, the k_0-only optimization in Phase 2 drove $k_{0,2}$ to 98.8% error because total current density is insensitive to R2—the optimizer minimizes the residual by effectively zeroing out the small R2 contribution. This poisoned Phases 3–5, which could not recover from the corrupted $k_{0,2}$ starting point.

12.2 v2 Pipeline (4-Phase) — Partial Success

The revised protocol removed the k_0-only phase and replaced it with a multi-observable joint optimization (total + peroxide current) starting from cold k_0 initial guesses. Phase 2 achieved R2 recovery: $k_{0,2}$ error 9.2%, α_2 8.5%. However, Phase 3 used total-current-only regularization annealing, which destroyed R2: $k_{0,2}$ diverged to 137% and α_2 to 90%. This confirmed the "never total-current-only" rule: *every* phase that jointly optimizes all four parameters must include the peroxide observable.

12.3 v3 Pipeline (3-Phase, Multi-Obs Throughout) — Final

The final protocol enforces multi-observable fitting in every joint optimization phase:

1. **Phase 1: α-only symmetric.** 12-point symmetric range, α-only optimization. Result: α_1 error = 0.36%.

2. **Phase 2: Multi-obs shallow cathodic.** 10-point shallow range ($\hat{\eta} \in [-1, -13]$), multi-observable joint (k_0, α) from cold k_0 start. Result: all four parameters 8–18% error.

3. **Phase 3: Multi-obs full cathodic.** 15-point full range ($\hat{\eta} \in [-1, -46.5]$), multi-observable joint, warm-start from Phase 2. Result: improved 3 of 4 parameters.

The shallow-then-full strategy avoids solver failures during early optimization (when parameters are far from truth) and adds the deep plateau constraint once parameters are in the correct basin.

Table 11: Master pipeline: complete version comparison.

Method	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
Staged S2 (15pt cath.)	4.9	98.5	0.1	72.3	98.5
Symmetric joint (20pt)	296.0	850.0	1.8	84.0	850.0
Multi-pH v2 (3 conds.)	6.5	95.8	9.2	90.0	95.8
Master v1 best	4.7	98.8	0.0	73.2	98.8
Master v2 best (P2)	18.1	9.2	9.2	8.5	18.1
Master v3 best (P3)	**16.7**	**6.1**	**8.2**	**11.7**	**16.7**

The final v3 result recovers all four BV parameters to within 17% (Table 11). In physical units, $k_{0,1} = 2.8 \times 10^{-8}$ m/s (true 2.4×10^{-8}) and $k_{0,2} = 1.1 \times 10^{-9}$ m/s (true 1.0×10^{-9}).

Three key findings emerge from the iterative development:

1. **Peroxide current is indispensable for R2.** No total-current-only method recovered $k_{0,2}$ or α_2. The peroxide observable provides the independent equation needed to disentangle R1 and R2.

2. **"Never total-current-only" rule.** Every phase that jointly optimizes all four parameters must include the peroxide observable. This was validated across v2 (Phase 3 diverged) and v3 (multi-obs Phase 3 improved).

3. **R1 vs. R2 trade-off.** Methods optimized for R1 achieve $k_{0,1} < 5\%$ but $k_{0,2} > 98\%$. Methods including the peroxide observable achieve $k_{0,2} < 10\%$ but $k_{0,1} \sim 17\%$. No method yet achieves $< 10\%$ on all four parameters simultaneously with 2% noise.

13 Speed Optimization of Master Pipeline

The v3 master pipeline (§12) was optimized for wall-clock time without sacrificing accuracy. Two changes were retained in v4: (i) shared forward solutions for multi-observable phases (the forward PDE solve is identical for total and peroxide current, so removing cache clearing between observables allows the checkpoint-restart fast path), and (ii) predictor validation in the warm-start solver (early revert for bad predictor states reduces wasted SNES iterations). Several more aggressive strategies—fewer voltage points, skipping Phase 1, and relaxed convergence tolerances—were tried and reverted because they degraded $k_{0,1}$ error from 16.7% to 52–58%. Further speedup ($> 15\%$) would require parallel MPI point-solves or GPU acceleration.

Table 12: v4 speed optimization: per-phase timing. All parameter errors within 0.03% of the v3 baseline.

Phase	v3 Time (s)	v4 Time (s)	Speedup (%)
P1 (α symmetric)	172	172	0.0
P2 (multi-obs shallow)	819	696	15.0
P3 (multi-obs full)	1084	964	11.1
Total	**2075**	**1832**	**11.7**

14 Parallelism Optimization: 4.4× Speedup

Three rounds of optimization were applied to the v4 master pipeline (1832 s baseline), exploiting the independence of voltage-point solves after the initial warm-start sweep populates the checkpoint cache.

Round 1 — Parallel Fast Path (v5, 2.0×). After the first sequential sweep, all subsequent L-BFGS-B evaluations solve each voltage point independently from cached initial conditions. A `ProcessPoolExecutor` with `spawn` context (matching the existing Robin pipeline in `FluxCurve/point_solve.py`) dispatches these solves to worker processes. A new module `FluxCurve/bv_parallel.` provides the worker infrastructure: serializable config, per-worker Firedrake initialization, and forward+adjoint computation from cached ICs. Phase 2 dropped from 696 s to 346 s and Phase 3 from 964 s to 408 s.

Round 2 — Multi-Observable Workers (v6, 3.5× cumulative). Workers were extended to compute *both* adjoint gradients (total and peroxide current) after a single forward solve, eliminating the sequential secondary-observable sweep. A "two-tape-pass" approach clears the adjoint tape after each observable, rebuilds forms, and loads the converged state for a single zero-residual SNES step before computing the adjoint. Phase 2 dropped to 168 s and Phase 3 to 188 s.

Round 3 — Optimizer Tolerances + Pool Reuse (v7, 4.4×). Relaxed L-BFGS-B convergence ($\texttt{gtol} = 5\times10^{-6}$, $\texttt{ftol} = 10^{-8}$) for Phases 2/3 reduced evaluation counts ($\sim$40 $\rightarrow$ $\sim$26 and $\sim$38 $\rightarrow$ $\sim$15, respectively). Reusing a single worker pool across phases saved $\sim$10 s of process-spawn overhead. Accuracy paradoxically *improved*, suggesting the tighter tolerances had caused overshooting.

The remaining 415 s is dominated by target generation ($\sim$70 s, sequential warm-start required), the first sweep per phase ($\sim$55 s, sequential), parallelized fast-path evaluations ($\sim$210 s at 9 workers), and `spawn`-context pool initialization ($\sim$40 s, irreducible with Firedrake import overhead). Further

Table 13: Parallelism optimization: timing and accuracy progression.

Version	P1 (s)	P2 (s)	P3 (s)	Total (s)	Speedup
v4 (serial)	172	696	964	1832	1.0
v5 (parallel)	167	346	408	923	2.0
v6 (multi-obs)	162	168	188	520	3.5
v7 (final)	162	135	117	415	4.4

Table 14: Accuracy improvement from v4 to v7 (parameter errors, %).

Param	v4 (%)	v7 (%)	Change (pp)
$k_{0,1}$	16.7	10.9	−5.8
$k_{0,2}$	6.1	2.6	−3.5
α_1	8.2	5.6	−2.6
α_2	11.7	8.8	−2.9

strategies (worker solver reuse, multi-fidelity mesh, speculative parallel first eval, MPI ensemble) were assessed and rejected as low-reward or architecturally incompatible.

15 POD/RBF Surrogate Model

A radial basis function (RBF) surrogate model was developed to replace PDE solves during inference, mapping BV parameters directly to I–V curve observables. Training data were generated by running the `FluxCurve` pipeline at Latin hypercube–sampled (LHS) parameter combinations, with k_0 sampled in $\log_{10}$ space and all inputs z-score normalized. The surrogate uses `scipy.interpolate.RBFInterpolator` with a thin-plate spline kernel and linear polynomial tail (degree 1). Gradients are computed via central finite differences (8 surrogate evaluations per gradient, total cost ∼microseconds).

Three rounds of development were performed. **Round 1 (v8)**: $N = 200$ LHS samples (181 valid, 90.5% convergence rate) at 22 voltage points; current-density NRMSE 0.06%, peroxide-current NRMSE 0.54%. **Round 2A (v8.1)**: removed a degraded full-cathodic surrogate phase, clamped optimizer bounds to the training data range, and added PDE refinement—which did not improve results because the surrogate acts as an implicit regularizer, smoothing out noise realizations that PDE solves faithfully reproduce. **Round 2B (v9)**: $N = 500$ LHS samples (445 valid) including 100 focused near the true parameters, with cross-validated smoothing (optimal: exact interpolation, $\epsilon = 0$).

Table 15 shows that, once the surrogate is trained (a one-time offline cost) and the target I–V curve is generated (∼70 s, shared by all methods), the actual parameter optimization takes **less than 0.1 s**—effectively instantaneous compared to ∼345 s for PDE-based v7 ($> 3000\times$ speedup on the optimization step alone). The previously reported "total time" of ∼70 s for v8/v9 was dominated entirely by target data generation; the surrogate optimization itself runs in milliseconds. The surrogate achieves competitive accuracy on $k_{0,1}$, α_1, and α_2. However, $k_{0,2}$ accuracy is worse (7.6–9.6% vs. 2.6% for PDE-based v7) because the peroxide-current surrogate has higher interpolation error (NRMSE 0.54–11.4%) than the current-density surrogate (0.06–0.42%). Increasing training data from 181 to 445 samples improved $k_{0,2}$ (9.6% → 7.6%) at the cost of ∼3× longer offline training (∼3.1 hr → ∼9.5 hr), with diminishing returns. The surrogate is most valuable when

Table 15: Comparison of PDE-based (v7) and surrogate-based (v8, v9) inference. Surrogate errors are from the best surrogate-only phase; PDE refinement from surrogate starting points did not improve accuracy. Target generation (∼70 s) is a shared fixed cost across all versions and is excluded from the optimization time. Offline training data generation is a one-time amortized cost.

Version	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Optim. time (s)	Offline cost
v7 (PDE)	10.9	2.6	5.6	8.8	345	—
v8 (Surr, 181)	7.3	9.6	4.5	4.6	< 0.1	∼3.1 hr
v9 (Surr, 445)	8.8	7.6	4.8	6.4	< 0.1	∼9.5 hr

rapid inference is needed (e.g., many repeated inference runs, real-time applications, or MCMC sampling).

16 Conclusions

- **The 3-phase master protocol v3 recovers all four BV parameters to within 17%**: $k_{0,1}$: 16.7%, $k_{0,2}$: 6.1%, α_1: 8.2%, α_2: 11.7%. This is the first method to achieve meaningful simultaneous recovery of both R1 and R2 kinetic parameters.
- **Parallelism optimization achieves a 4.4× speedup (1832 s → 415 s) with improved accuracy**: process-level parallelism of independent voltage-point solves, multi-observable workers, relaxed optimizer tolerances, and pool reuse reduce wall-clock time from ∼30.5 min to ∼6.9 min. All four parameter errors improved by 2.6–5.8 percentage points.
- **Multi-observable (total + peroxide current) is indispensable for R2 identifiability**: no total-current-only method recovered $k_{0,2}$ or α_2. Fitting both observables simultaneously reduces $k_{0,2}$ error from ∼97–100% to 6.1% and α_2 from ∼67–90% to 11.7%.
- **Pipeline ordering is critical—"never total-current-only"**: every phase that jointly optimizes all four parameters must include the peroxide observable. k_0-only optimization on total current drives $k_{0,2}$ to near-zero (v1), and total-current-only regularization annealing destroys R2 (v2 Phase 3).
- **An R1 vs. R2 accuracy trade-off exists**: methods optimized for R1 achieve $k_{0,1} < 5\%$ but $k_{0,2} > 98\%$; methods including the peroxide observable achieve $k_{0,2} < 10\%$ but $k_{0,1} \sim 17\%$. No method yet achieves $< 10\%$ on all four parameters with 2% noise.
- **Shallow-then-full multi-obs is the best architecture**: Phase 2 (shallow $[-1, -13]$) avoids solver failures during early optimization; Phase 3 (full $[-1, -46.5]$) adds the deep plateau constraint once parameters are in the correct basin.
- **Symmetric (anodic + cathodic) data achieves the best α_1 recovery**: 1.8% error via direct joint inference, confirming the physics prediction about the anodic Tafel slope.
- **Forward solver robustness was substantially improved**: mesh refinement ($N_y = 300$, $\beta = 4.0$), smaller timesteps ($\Delta t = 0.1$), denser voltage sweeps (19 points, max_eta_gap = 1.5), softplus regularization, and SNES linesearch tuning raised target generation convergence from 78% to 100%.

- **Multi-pH improves R1 but is limited by solver convergence**: sharing (k_0, α) across pH conditions yields excellent R1 ($k_{0,1}$: 6.5%, α_1: 3.6%) and partial α_2 improvement (29.5%), but the charged solver fails at elevated H^+ concentrations.

- **Regularization helps but anchors to the prior**: for $\lambda \geq 0.001$, recovered parameters reflect the prior rather than the truth.

- **Bridge points and two-branch sweeps work robustly**: zero bridge failures across 2046+ evaluations; two-branch sweep enables anodic data collection with hub-state warm-start.

- **The RBF surrogate reduces optimization time from $\sim$345 s to < 0.1 s ($> 3000\times$) with complementary accuracy trade-offs**: once the surrogate is trained (one-time cost of $\sim$3–10 hr) and target data are generated ($\sim$70 s, shared by all methods), the actual parameter optimization is essentially instantaneous. The surrogate improves $k_{0,1}$ (7.3% vs. 10.9%), α_1 (4.5% vs. 5.6%), and α_2 (4.6% vs. 8.8%), but degrades $k_{0,2}$ (9.6% vs. 2.6%) due to higher peroxide-current interpolation error.

- **The surrogate acts as an implicit regularizer**: by smoothing out noise realizations that PDE solves faithfully reproduce, the surrogate objective landscape is smoother. PDE refinement from surrogate-recovered parameters does not improve (and can worsen) accuracy, because the PDE solver fits a different noise realization.

- **Surrogate accuracy has diminishing returns**: increasing training data from 181 to 445 samples improved $k_{0,2}$ by only 2 percentage points (9.6% $\rightarrow$ 7.6%) at 3$\times$ one-time offline training cost ($\sim$3.1 hr $\rightarrow$ $\sim$9.5 hr; amortized over all subsequent inference runs). The PDE-based v7 pipeline remains the best option when $k_{0,2}$ accuracy is the priority.

17 Next Steps

Completed

1. **Multi-observable joint fitting.** COMPLETED. Fitting total current and peroxide current simultaneously achieved the R2 breakthrough: $k_{0,2}$ error 6.1%, α_2 error 11.7%.

2. **Multi-pH inference.** COMPLETED. Three-condition fitting with shared (k_0, α) achieved excellent R1 recovery and partial R2 improvement.

3. **Master inference protocol v1/v2/v3.** COMPLETED. Three iterations of pipeline design converged on a 3-phase multi-obs-throughout architecture that recovers all four BV parameters to within 17%.

4. **Forward solver robustness.** COMPLETED. Mesh refinement, smaller timesteps, denser voltage sweeps, softplus regularization, and SNES linesearch tuning raised target generation convergence from 78% to 100%.

5. **Speed optimization of master pipeline.** COMPLETED. Shared forward solutions and predictor validation achieved 11.7% wall-clock speedup (2075 s $\rightarrow$ 1832 s) with identical accuracy. More aggressive strategies were tried and reverted.

6. **Parallelism optimization (v5–v7).** COMPLETED. Process-level parallel point-solves, multi-observable workers, relaxed optimizer tolerances, and pool reuse achieved a 4.4$\times$ speedup

(1832 s $\rightarrow$ 415 s) with improved accuracy (all four errors decreased by 2.6–5.8 percentage points).

7. **POD/RBF surrogate model (v8–v9).** COMPLETED. RBF thin-plate spline surrogate trained on 181–445 LHS samples reduces optimization time from $\sim$345 s to < 0.1 s ($> 3000\times$ speedup), with competitive accuracy on three of four parameters. Target generation ($\sim$70 s) is a shared fixed cost; offline training ($\sim$3–10 hr) is a one-time amortized cost. Diminishing returns observed: increasing training data $3\times$ improved $k_{0,2}$ by only 2 percentage points. The surrogate provides a complementary approach to PDE-based inference; the PDE pipeline remains best for $k_{0,2}$ accuracy (2.6% vs. 7.6%).

Future work

8. **Combined multi-observable + multi-pH + symmetric.** Multi-observable breaks R2 degeneracy; multi-pH adds concentration diversity; symmetric data constrains α. Combining all three in a single pipeline could push the R1 vs. R2 trade-off frontier below the current 17% ceiling.

9. **Neutral-species formulation for wider pH range.** The charged solver fails at $c_{\mathrm{H}^+} > 0.3$ due to Debye-length stiffness. A neutral-species formulation would avoid Poisson stiffness entirely, enabling multi-pH inference across the full experimentally relevant range.

10. **Gauss–Newton optimizer.** Preconditioning the gradient to decorrelate k_0 and α could reduce iteration counts and improve conditioning of the joint objective.

11. **Experimental data fitting.** Apply the validated v7 pipeline to real RRDE data from Mangan et al. (2025), which provides both disk (total) and ring (peroxide) currents. The surrogate could accelerate initial parameter sweeps before PDE-based refinement.

12. **Bayesian uncertainty quantification.** Replace point estimates with posterior distributions via ensemble Kalman inversion (EKI) or MCMC to characterize parameter uncertainty and the R1–R2 correlation structure. The surrogate's microsecond evaluation time makes MCMC sampling practical.

13. **Transient (chronoamperometric) data.** R1 and R2 reach steady state on different timescales; fitting $I(t)$ provides temporal separation invisible in steady-state data.

14. **MPI Ensemble parallelism for further speedup.** The current `ProcessPoolExecutor` approach is near its ceiling (9 workers, $\sim$40 s pool init overhead). MPI Ensemble parallelism would require restructuring from a single-process to an `mpiexec` launcher; deferred to future work.